



FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065 (INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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GITHUB LINK:	https://github.com/brryes/Reyes-SofiaBeatriceAlexa-CS0065-AN42.git

I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
(Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF file**.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
 - Classes and objects
 - If-Else statements
 - Loops (for loop)
 - Functions / methods
2. Understand the environment:
 - Two rooms: A and B
 - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output

```
C:\Users\hazel\AppData\Local\Programs\Python\Python39\python.exe C:\Users\hazel\AppData\Local\Programs\Python\Python39\python.exe
Is room A Dirty or Clean? dirty
Is room B Dirty or Clean? clean

How many steps should the agent run? 5

Step 1: Agent is in room A
Room A has been cleaned.
Environment state: {'A': 'Clean', 'B': 'Clean'}

Step 2: Agent is in room A
Moving to room B
Environment state: {'A': 'Clean', 'B': 'Clean'}

Step 3: Agent is in room B
Moving to room A
Environment state: {'A': 'Clean', 'B': 'Clean'}

Step 4: Agent is in room A
Moving to room B
Environment state: {'A': 'Clean', 'B': 'Clean'}

Step 5: Agent is in room B
Moving to room A
Environment state: {'A': 'Clean', 'B': 'Clean'}
Press any key to continue . . .
```

Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:

- Rule 1: If the current room is dirty, clean it.
 - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
 - The agent's current location
 - The action taken (cleaned/moved)
 - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)

```
C:\Users\hazel\AppData\Local\Programs\Python\Python38-32\python.exe
Is room B Dirty or Clean? dirty
Is room C Dirty or Clean? dirty

How many steps should the agent run? 5

Step 1:
[A-Agent] [B-Dirty] [C-Dirty]
Moving to room B
[A-Clean] [B-Agent] [C-Dirty]

Step 2:
[A-Clean] [B-Agent] [C-Dirty]
Room B has been cleaned.
[A-Clean] [B-Agent] [C-Dirty]

Step 3:
[A-Clean] [B-Agent] [C-Dirty]
Moving to room C
[A-Clean] [B-Clean] [C-Agent]

Step 4:
[A-Clean] [B-Clean] [C-Agent]
Room C has been cleaned.
[A-Clean] [B-Clean] [C-Agent]

Step 5:
[A-Clean] [B-Clean] [C-Agent]
All rooms are clean. Agent is idle.
[A-Clean] [B-Clean] [C-Agent]
Press any key to continue . . .
```

Your Code - Laboratory Tasks

```
# Rule-Based Vacuum Cleaner Agent
```

```

# Task 1: Set up the environment
rooms = {
    "A": input("Enter the condition of Room A (dirty/clean): ").lower(),
    "B": input("Enter the condition of Room B (dirty/clean): ").lower()
}

# Check room conditions
def is_dirty(room):
    return rooms[room] == "dirty"

# Clean a room
def clean_room(room):
    rooms[room] = "clean"

# Task 2: Create the rule-based agent
class VacuumAgent:

    def __init__(self):
        self.current_room = "A"

    # Move between rooms
    def move(self):
        if self.current_room == "A":
            self.current_room = "B"
        else:
            self.current_room = "A"

    # Follow the rules
    def perceive_and_act(self):
        if is_dirty(self.current_room):
            clean_room(self.current_room)
            return "Cleaned the room"
        else:
            self.move()
            return "Moved to Room " + self.current_room

```

```
# Task 3: Run the simulation
agent = VacuumAgent()

steps = int(input("Enter the number of steps: "))

print("\nInitial Environment:")
print(rooms)

for step in range(1, steps + 1):
    action = agent.perceive_and_act()

    print("\nStep", step)
    print("Agent location:", agent.current_room)
    print("Action:", action)
    print("Environment:", rooms)

print("\nSimulation complete!")
```

```
le_based_agent.py
Step 1
Agent location: A
Action: Cleaned the room
Environment: {'A': 'clean', 'B': 'clean'}

Step 2
Agent location: B
Action: Moved to Room B
Environment: {'A': 'clean', 'B': 'clean'}

Step 3
Agent location: A
Action: Moved to Room A
Environment: {'A': 'clean', 'B': 'clean'}

Step 4
Agent location: B
Action: Moved to Room B
Environment: {'A': 'clean', 'B': 'clean'}

Step 5
Agent location: A
Action: Moved to Room A
Environment: {'A': 'clean', 'B': 'clean'}

Simulation complete!
PS C:\Users\bea\Documents\elective\ta2> |
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

MY ANSWERS:

1. The agent cleans the room if it is dirty. If the room is clean, it moves to the other room.
2. It checks the condition of its current room. It cleans when the room is dirty and moves when the room is already clean.
3. If both rooms are clean, the agent will move between Room A and Room B without cleaning anything.
4. I would store more rooms in the dictionary and update the move method so the agent moves through all rooms in order.

